

```
dua_ansari@DESKTOP-4DF8PF8:~$ cd xv6-labs-2025
dua_ansari@DESKTOP-4DF8PF8:~/xv6-labs-2025$ make
make: 'kernel/kernel' is up to date.
dua_ansari@DESKTOP-4DF8PF8:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_Memdump user/_find user/_exec user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1144 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep 100
$ sleep 100
```

```
dua_ansari@DESKTOP-4DF8PF8:~$ cd xv6-labs-2025
dua_ansari@DESKTOP-4DF8PF8:~/xv6-labs-2025$ make
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egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sleep 100
$ sleep 100
$ sixfive sixfive.txt
5
100
18
6
$ |
```

```
dua_ansari@DESKTOP-4DF8P X + v - □ X
/_sleep user/_sixfive user/_memdump user/_find user/_exec user/_uptime
user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1)
blocks 1953 total 2000
balloc: first 1182 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 12
8M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=
fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=v
irtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
B
1996488704
29295
l
Unknown format:
Example 5:
hello
world
$ |
```

```
dua_ansari@DESKTOP-4DF8P  X  +  v
riscv64-unknown-elf-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/_memdump.sym
mkfs/mkfs fs.img README user/_findtest.sh user/_sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep
user/_sixfive user/_Memdump user/_find user/_exec user/_memdump
nm -meta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1144 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ _memdump
25000
20
Z
hello xv6
$ echo > b
$ mkdir a
$ echo a/b
a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/aa/b
```

```
dua_ansari@DESKTOP-4DF8PF8: ~  
got:  
4  
expected:  
3  
QEMU output saved to xv6.out.test_find_sh  
== Test time ==  
time: FAIL  
Cannot read time.txt  
Score: 70/131  
make: *** [Makefile:363: grade] Error 1  
dua_ansari@DESKTOP-4DF8PF8: ~/xv6-labs-2025$ make  
make: 'kernel/kernel' is up to date.  
dua_ansari@DESKTOP-4DF8PF8: ~/xv6-labs-2025$ make qemu  
mv -f fs.img fs.img.bk  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_exec user/_uptime user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1)  
blocks 1953 total 2000  
ballocc: first 1200 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 2 starting  
hart 1 starting  
init: starting sh  
$ find . wc -exec echo hi  
find . wc -exec echo hi  
hi  
$ |
```

```
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (6.7s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (1.5s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: FAIL (2.5s)
...
    hart 1 starting
    init: starting sh
GOOD $ sleep 10
    $ echo FAIL
BAD  FAIL
    $ qemu-system-riscv64: terminating on signal 15 from pid 1929 (
make)
    unexpected lines in output
    QEMU output saved to xv6.out.sleep
```

```

dua_ansari@DESKTOP-4DF8P1 X + v - □ X
    hart 1 starting
    init: starting sh
GOOD $ sleep 10
    $ echo FAIL
BAD  FAIL
    $ qemu-system-riscv64: terminating on signal 15 from pid 1929 (
make)
    unexpected lines in output
    QEMU output saved to xv6.out.sleep
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (10.3s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: FAIL
    ...
        0
        0
GOOD 6
        0
        0
    ...
        0
        0
        0
        0
    $ qemu-system-riscv64: terminating on signal 15 from pid 1997 (
make)

```

```

dua_ansari@DESKTOP-4DF8P1 X + v - □ X
    QEMU output saved to xv6.out.sixfive_readme
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: FAIL (1.5s)
    ...
        init: starting sh
        $ sixfive sixfive.txt README
GOOD 5
GOOD 100
GOOD 18
GOOD 6
    $ qemu-system-riscv64: terminating on signal 15 from pid 2031 (
make)

```

```
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.5s)
(Old xv6.out.memdump_examples failure log removed)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.5s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.6s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.6s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (2.3s)
```

QEMU output saved to xv6.out.memdump_examples

```
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.5s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (10.6s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK
```

```
dua_ansari@DESKTOP-4DF8PF8: ~$ make qemu-gdb
find, recursive: OK (2.3s)
== Test exec ==
$ make qemu-gdb
exec: FAIL (1.6s)
...
    hart 1 starting
    init: starting sh
    $ find . wc -exec echo
    find . wc -exec echo
    $ qemu-system-riscv64: terminating on signal 15 from pid 22617
(make)
MISSING '^./wc$'
QEMU output saved to xv6.out.find_exec
== Test exec, multiple args ==
$ make qemu-gdb
Timeout! exec, multiple args: FAIL (30.4s)
    Number of appearances of 'hello'
    got:
    2
    expected:
    5
    QEMU output saved to xv6.out.find_exec_args
== Test exec, recursive find ==
$ make qemu-gdb
Timeout! exec, recursive find: FAIL (30.3s)
    Number of appearances of 'hello'
    got:
    4
    expected:
    3
    QEMU output saved to xv6.out.test_find_sh
== Test time ==
time: FAIL
    Cannot read time.txt
Score: 70/131
make: *** [Makefile:363: grade] Error 1
dua_ansari@DESKTOP-4DF8PF8: ~/xv6-labs-2025$ |
```